

Bird Abundance

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1. Set up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(dplyr)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
library(flextable)
library(ggribes)
library(NatParksPalettes)
```

Data

```
birds <- read_csv(here("data", "birds.csv"))

venoco_water <- read_csv(here("data", "venoco-water.csv"))
```

2. Cleaning

Wrangling birds

```
#created a new object from birds
bird_wy2025 <- birds %>%
  # exclude repeat observations
  mutate(observation_date = as.Date(observation_date, format = "%m/%d/%y"))
  ↪ %>%
  filter(observation_date > as.Date("2024-11-13") & observation_date <
  ↪ as.Date("2025-10-22"),
        repeat_observation == "No",
        e_bird_group %in% c("Waterfowl",
                           "New World Sparrows",
                           "Martins and Swallows",
                           "Shorebirds")) %>%

# filling in taxonomic information about order and families for eBird
↪ groups
mutate(lowest_group = case_when(
  e_bird_group == "New World Sparrows" ~ "Passerellidae",
  e_bird_group == "Martins and Swallows" ~ "Hirundinidae",
  e_bird_group == "Shorebirds" ~ "Charadriiformes",
  e_bird_group == "Waterfowl" ~ "Anatidae"
))

# creating new object from most_abundant_groups
abundant_species_counts <- bird_wy2025 %>%
  # grouping by speices
  group_by(season, e_bird_group, species) %>%
  # calculating total counts for each species
  summarize(total_count = sum(count)) %>%
  # arranging in descending order
  arrange(-total_count)
```

Wrangling Water

```
# Creating a new object from venoco_water
venoco_clean <- venoco_water |>
  # filter to only include 2025 water year
  filter(wy == "2025") |>
  # creating a new column to convert water elevation in ft to meters
  mutate(comp_level_m = (comp_level_ft + 2.96) * 0.3048) |>
  # selecting columns of interest
  select(date, comp_level_m, temperature)

# create a new object from Venoco_clean
venoco_daily <- venoco_clean |>
  # group by date
  group_by(date) |>
  # calculate mean water level for each day
  summarize(mean_level_m = mean(comp_level_m, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup()
```

Shorebird Wrangling

```
shorebird_total_count <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(lowest_group == "Charadriiformes") |>
  #group by species
  group_by(species) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  #arrange in descending order by total counts
  arrange(-total_count)
```

Wrangling Shorebird and Waterfowl

```
shorebird_waterfowl_25 <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
```

```

filter(e_bird_group %in% c("Waterfowl", "Shorebirds")) |>
#group by species
group_by(e_bird_group, observation_date) |>
#calc total counts per species
summarize(total_count = sum(count)) |>
ungroup() |>
left_join(venoco_daily,
# naming shared columns between both data frames
by = c("observation_date" = "date"))

```

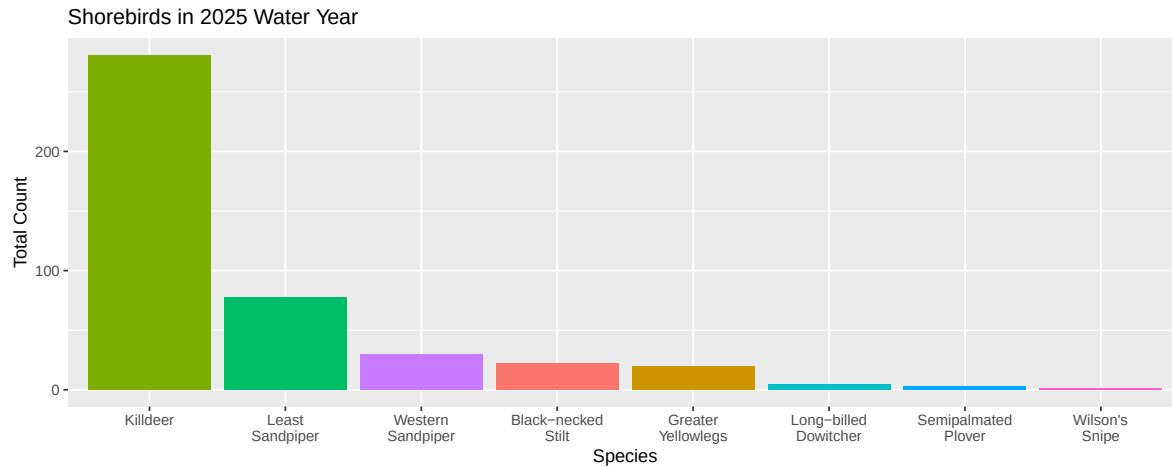
3. Exploring Visualizations

Plotting Shorebirds

```

ggplot(data = shorebird_total_count,
# x axis is species in descending order of total count
# # y axis is total count
# filling columns by species
mapping = aes(x = reorder(species, -total_count),
               y = total_count,
               fill = species)) +
#first layer: column to represent counts
geom_col() +
# taking out the legend
theme(legend.position = "none") +
#relabelling x and y axes, adding title
labs(x = "Species",
      y = "Total Count",
      title = "Shorebirds in 2025 Water Year") +
scale_x_discrete(labels = ~ str_wrap(., width = 10))

```



Plotting Shorebirds and Waterfowl counts over different Water Elevations

```
ggplot(data = shorebird_waterfowl_25,
       mapping = aes(x = mean_level_m,
                     y = total_count,
                     color = e_bird_group)) +
  geom_point() +
  geom_line() +
  theme_bw() +
  scale_color_manual(values = natparks.pals("Yellowstone")) +
  labs(x = "Venoco Bridge Water Elevation (meters)",
       y = "Bird Count",
       color = "Taxonomic Group",
       title = "Water Elevation affects on Shorebirds and Waterfowl in Water
↪ Year 2025")
```

